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Late Permian to Early Triassic magnetostratigraphy

Maja Haag and Friedrich Heller
Institut für Geophysik, ETH-Hönggerberg, CH-8093 Zürich, Switzerland

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ABSTRACT

A Late Permian to Early Triassic magnetostratigraphic reference section is presented. The Lower Triassic part is based on results from marine limestone sections in South China published earlier [1,2]. Reliable new Permian data are added here which have been collected in the Nammal gorge (Salt Range, Northwest Pakistan) where marine sediments have been deposited quasi-continuously with occasional minor hiatuses during the late Palaeozoic to early Mesozoic.

About 50% of the Permian samples from the Nammal section contain, hidden beneath a strong recent or Tertiary overprint, a characteristic remanent magnetization (ChRM) which is very likely of Permian age. This component, which was imprinted on the southern hemisphere, has normal as well as reversed polarity with a normal mean direction (Decl. = 289.3°, Incl. = −50.3°, $\alpha_{95} = 4.3°$, $N = 113$) which is in close agreement with the palaeofield direction expected for a site belonging to the Indian plate as part of Gondwanaland during the Permian.

In the lower Upper Permian several normal polarity zones are recognized. This contradicts the current assumption that rocks of this age belong to the long, reversely polarized Kiaman hyperzone. The Kiaman interval must end and the Illawarra hyperzone of mixed polarity must begin in or prior to the lowermost Upper Permian. The Permian/Triassic boundary at Nammal as well as in the Chinese sections is situated very close to a transition from a reversed to a normal polarity zone. The Upper Permian at Nammal together with the Lower Triassic South China sections is estimated to cover about 20 Ma. Nearly 30 polarity changes are observed which result in an average reversal frequency very similar to that observed during the early Tertiary. The reversal rate after the end of the long-lasting reversed Kiaman hyperchron apparently increases in a manner similar to that after the end of the Cretaceous Long Normal Superchron. Only a few polarity zones are found in the lower Upper Permian, while progressively more frequent reversals follow in the Lower Triassic.

1. Introduction

Towards the end of the Permian, and specifically at the Permian/Triassic (P/Tr) boundary, the most spectacular Phanerozoic extinction of life on earth took place [3]. Approximately half of all families in the marine environment disappeared and vertebrates on land were decimated. The "great dying" in the Late Permian probably lasted a million years, but reached its climax towards the very end of the Permian [4] possibly due to sea level changes [5]. Correlation and synchronism of the marine and non-marine records of the events at the end of the Palaeozoic is difficult to ascertain, as with other extinction events (e.g. [6]). The boundary is positioned in the marine environment at the end of the Dorashamian, whereas the Tatarian/Scythian boundary determines the transition in continental sediments.

The polarity reversal history of the earth's magnetic field has important potential for dating because of the worldwide synchronous occurrence of reversals. The reversal history in the Late Palaeozoic, however, is not known yet and magnetostratigraphic studies in sediments older than Middle Jurassic lack the possibility of comparison with oceanic magnetic anomalies. In addition, the geological record of many P/Tr boundary sections suffers from inaccurately known sedimentary history and deposition rates. For this reason the earlier magnetostratigraphic work mostly done in unfossiliferous continental sediments [7–9] can barely satisfy minimum criteria for a magnetostratigraphic Permian/Triassic reference section.

Palaeontologically well-dated marine sections where youngest Permian rocks are overlain by oldest Triassic sediments without a marked discontinuity are rare but have been found, devel-

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oped mainly in the Tethyan realm [10]: in Kashmir, Armenia (Dzhulfa region), Northwest Iran, Central Iran (Abadah region) and South China, and with some restrictions in the Salt Range (Northwest Pakistan). Well-developed P/Tr boundary sections are also found in East Greenland (Kap Stosch region).

Earlier studies [7,8,11,12] have provided evidence that the earth's magnetic field had constant reversed polarity for about 60 Ma between the Upper Carboniferous (Early Stephanian) and the Upper Permian to Lower Triassic. This long-lasting polarity period was named the Kiaman interval after the early discovery of reversely magnetized basalts at Kiama-Wollogong (New South Wales) [13]. The Kiaman in Australia is followed by a normal polarity zone (Lower Triassic Narrabeen sediments [14]) and subsequently by a zone of mixed polarity which together have been termed the Illawarra reversals [11]. The Illawarra superchron covers mainly the Lower Triassic and is thought to start in the uppermost Permian. This onset of the Illawarra reversals is supported by recent results from North America [15] which suggest that the Kiaman interval had ended prior to 251 ± 4 Ma ago in the uppermost Permian according to the timescale by Haq and Van Eysinga [16].

In this study a magnetostratigraphic profile is described which represents one of the most complete Upper Permian marine stratigraphic sections. Suitable rock magnetic characteristics and well-determined biostratigraphic control allow for the development of a magnetostratigraphic reference section during the Late Permian. The magnetostratigraphy can be extended across the Permian/Triassic boundary by combination with Early Triassic results from South China [1,2,17].

2. Sampling, lithology and stratigraphy

The P/Tr boundary is exposed in different sections of the Salt Range (Fig. 1) and of the Transindus Ranges in Northern Pakistan. The most complete Upper Permian to Lower Triassic section is exposed in the Nammal gorge (32.6° N, 71.8° E). Nodular limestones, dolostones and marls which were formed in a shallow marine environment represent the Upper Permian Wargal formation and the lower part of the Upper Permian to Lower Triassic Kathwai member of the Mianwali formation (Table 1). The intervening Chhidru formation is made up of white to red sandstones of uniform grain size, bioclastic sandstones of heterogeneous grain size and sandy limestones which reflect a nearshore environment. Stratigraphical and palaeontological studies have been summarized by Kummel and Teichert [18] and by the

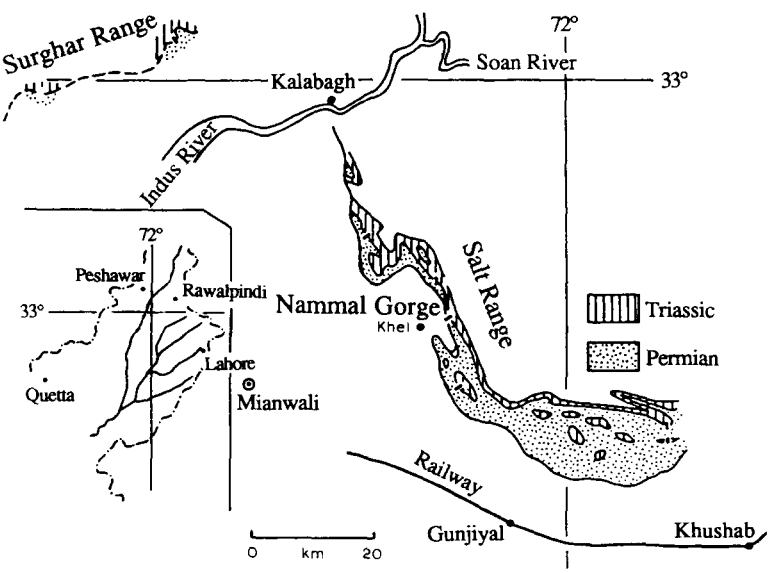


Fig. 1. Geographical location of the Permian and Triassic sediments in the Salt Range (adapted from [16]).

Pakistani–Japanese Research Group [19]. The tectonic situation has been reviewed by Powell [20].

More than 220 cores were drilled across the 200 m thick Upper Permian sediments of the Wargal and Chhidru formation, which are tilted towards the northeast with variable dip between 21° and 49°. Only the first 1.05 m of the Upper Permian/Lower Triassic Kathwai member were sampled. The remainder of the Triassic part of the section is being examined by H. Feinberg (Paris).

Early investigations [21,22] placed the P/Tr boundary at the top of the Kathwai member of the Mianwali formation (Table 1, arrow at level 1) where the largest faunal change occurs. Kummel and Teichert [18] took the abrupt lithology change between the white sandstone of the uppermost Chhidru formation and the dolomite units of the bottom Kathwai member as a boundary indicator (Table 1, arrow at level 3). These authors argued that the abrupt change from regressive (white sandstone) to transgressive lithology (dolomite) in the Salt Range, which is marked by a several centimetre thick black clay horizon, indicates a stratigraphic gap between the uppermost Permian and the lowermost Triassic. The most detailed and thorough investigation was completed by the Pakistani–Japanese Research Group (PJRG [19]) in 1985. They divided the two and a half metre thick Kathwai member of the lower Mianwali formation into three units: (1) a lower dolostone and sandstone unit of about 20 cm in thickness at Nammal with uppermost Permian brachiopods, (2) a middle or main dolomite unit whose age is characterized by the first appearance of *Ophiceras*, but is probably missing some parts of the *Otoceras woodwardi* zone which is considered to represent the oldest Triassic division [23], and (3) an upper dolomite and limestone unit with Ophiceratid ammonoids. The modern location of the boundary is then placed between the lower and middle unit of the Kathwai member (Table 1, arrow at level 2). The underlying Chhidru formation has been assigned a late Dzhulfian age. No clear palaeonto-

TABLE 1

Schematic Upper Permian to Lower Triassic chronostratigraphical division in the USA and Eurasia according to Haq and Van Eysinga [16] and Baud et al. [5] compared to lithostratigraphy and stratigraphical association of the corresponding Salt Range sediments according to Kummel and Teichert [10] and the PJRG [19]. P/Tr boundary (arrows): (1) Waagen [21] and Gee [22]; (2) PJRG [19]; (3) Kummel and Teichert [18]

Time [Ma]	Haq &Van Eysinga (1987)		Baud et al. (1989)	Kummel & Teichert (1973)		PJRG (1985)		
Tr_m	USA	Eurasia (general)	Eurasia	Salt Range		Salt Range		
244 Tr_l	Spathian	Scythian	Olenikian	Scythian	Mianwali Fm.	Narmia Member	Mianwali Fm.	
	Smithian					Mittiwali Member		
	Dienerian		Induan					<--- 1
Tr	Griesbachian					Kathwai Member		<--- 2
250 P	Ochoan	Tatarian	Dorashamian (Changxing.)	Chhidruan	Chhidru Fm.			<--- 3
						? Gap ?		
			Dzhulfian			Chhidru Fm.		
			Midian					
258	Guadalupian	Kazanian	Murgabian	Guadalupian	Wargal Fm.	Wargal Fm.		
261 P_u		Kungurian				Amb Fm.		
270 P_l	Leonardian	Artinskian		Artinskian	Amb Fm.			

logical evidence for latest Permian Dorashamian/Changxingian age has been found, although different brachiopod species may represent this age. The underlying Wargal formation extends from the upper Murgabian (*Neoschwagerina margaritae* zone in Tethys) into the lower Dzhulfian.

3. Magnetic measurements

The 1 inch samples were measured with a 3-axis ScT cryogenic magnetometer [24] which has a noise level of 5×10^{-6} A m^{-1}. In order to obtain well-defined magnetization directions, usually ten readings per measurement along each of the magnetometer axes were integrated in each of three different sample positions. Pilot samples were demagnetized stepwise in alternating fields (AF) of 5–60 mT as well as thermally from 100 °C to 700 °C. AF treatment produced substantial intensity loss of natural remanent magnetization (NRM) but no significant separation of NRM components until 60 mT. Therefore, most samples were treated thermally, the maximum demagnetization temperature depending on the sample magnetic mineralogy. The magnetization of many samples fell below the noise level of the cryogenic magnetometer at temperatures above 450 °C. At about the same temperature severe mineral changes were observed, often as reflected by a drastic increase in low field susceptibility. The newly generated magnetic minerals are associated with new laboratory induced magnetizations which were probably acquired in the 10 nT residual field of the cooling chamber. They were strong enough to inhibit investigation of the remaining original magnetization at elevated temperatures. Therefore it was usually necessary to stop thermal demagnetization between 400 and 500 °C.

NRM intensities of the grey marine carbonates and marls are weak; the mean value is 9×10^{-5} A m^{-1} (Fig. 2). The sediments mostly have very low concentration of ferromagnetic minerals, which is also reflected by low initial susceptibilities. About 40% of all 326 specimens have a diamagnetic susceptibility and another 22% have extremely weak susceptibility below 1×10^{-5} S.I. units. NRM directions before bedding correction preferentially plot with northern declinations and inclinations around 50 ° (Fig. 2), being aligned more or less parallel to the direction of the present axial

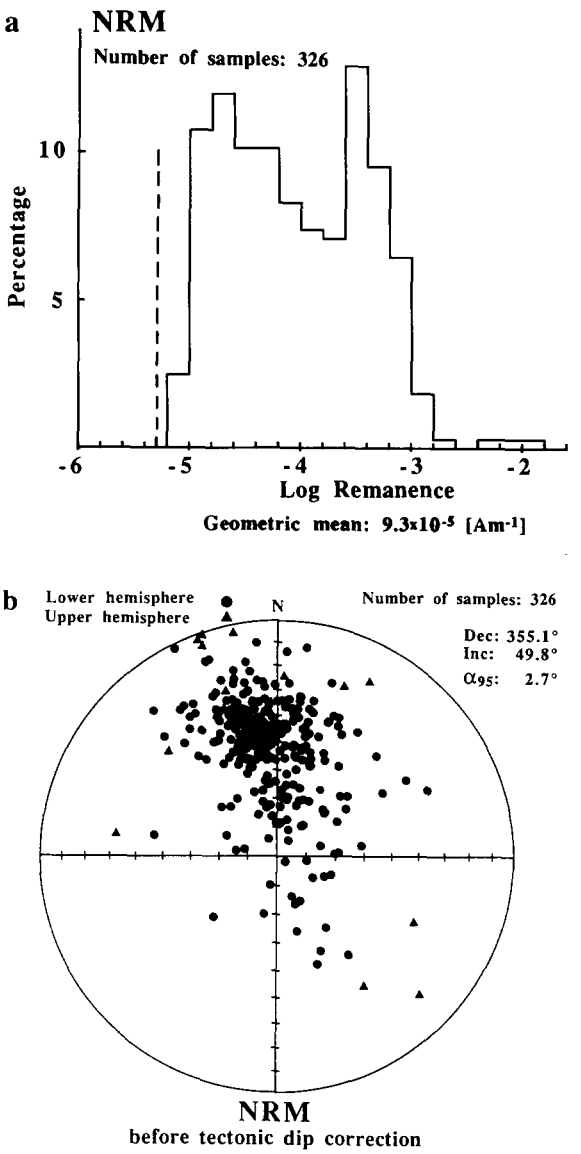


Fig. 2. (a) Histogram of natural remanent magnetization (NRM) intensities. The dashed line marks the sensitivity limit of the cryogenic magnetometer. (b) Stereographic projection of the in-situ NRM directions which concentrate around the local present-day field direction.

dipole field in Northern Pakistan (inclination $I = 52°$).

Three directional components of the NRM vectors can be resolved by thermal demagnetization. Some samples (Fig. 3a) show only the already mentioned present-day field component with northerly declination and an inclination of approximately +50 ° defined by a linear segment in

the orthogonal vector plot between 20 °C and 300 °C. In a second group of samples (Fig. 3b), NRM components are removed between 50 °C and 300–350 °C, which are given by straight-line segments with again northerly declination before bedding correction, but with distinctly shallower mean inclination of +19°. After bedding correction the declination remains northerly but the inclination becomes nearly horizontal. This shallow inclination component, like the present-day field component, often extends into the highest measurable blocking temperatures ranges. In 50% of the samples, however, we observe superimposed on either of these components a third component which is characterized by well-defined linear segments between 300 °C and 450–550 °C. It has an *in-situ* northwesterly declination and a negative inclination of about −30° (Fig. 3c). If there are mineral changes above 400–450 °C this third magnetization component is defined by the stable endpoint direction between 250 and 450 °C (Fig. 3d). After bedding correction the declination remains northwest but the inclination steepens to about −50°. This component occurs in both polarity states, i.e. also with southeasterly declination and relatively steep positive inclination (Fig. 3d and 4).

The shallow inclination component (second group) is found mainly in normal polarity state (Fig. 4). If imprinted before tilting, this magnetization must have been acquired at low northern latitudes. Secondary NRM components with low to intermediate blocking temperatures are imprinted often during tectonic events (e.g. [25]). This magnetization possibly records the timing and place of the collision of Pakistan/India with first the Ladakh island arc and second the Eurasian continent during the Palaeocene to Oligocene [26,27]. The pole calculated from this Tertiary mean direction (Table 2) has been plotted on the apparent polar wander path of Indo-Pakistan [26], where no distinction had been made between the rotation of the Salt Range and the general motion of the India/Pakistan subcontinent (Fig. 5). It is in agreement with the published palaeomagnetic data of the same age from the Salt Range, the Siwaliks and India. A similar Tertiary overprint has been found in many Palaeozoic to Tertiary rocks from the Indopakistan plate [28–30]. The wide scatter in declination possibly indicates that the magnetization was acquired partly during the rotation of the Salt Range. If a sample had both, a present earth field and the third magnetization component, then this shallow inclination component may possibly be the resultant of these two directions. However, not one sample showed all three components. The samples had either the present-day field or this shallow inclination component together with the third magnetization. Additionally, the well-defined straight-line segments of this shallow inclination component preclude a composite magnetization vector.

About half of the 232 demagnetized Nammal samples contain the third magnetization component with steeper inclinations around 50° in both normal or reversed polarity state (Fig. 4). This component is considered as the original, probably Permian characteristic magnetization (ChRM). Due to unremovable secondary components some directional scatter remains for this component. Many samples show only trends towards the normal or reversed mean direction. Because secondary components still contribute over the whole range of measurable demagnetization stages they have not been accepted as being representative of the Permian geomagnetic field. Only samples with absolute inclination values after bedding correction $> 20°$ (the median inclination between the expected Tertiary and Permian direction) have been taken into account in Fig. 4 and the magnetostratigraphic profile (Fig. 7). After application of this rejection criterion the resulting normal mean direction has a declination $D = 289°$ E and an fairly steep inclination $I = -50°$ (after bedding correction).

The bedding dips, which vary between 21° and 49° along the section, can be used for an attitude test. The attitude correction improves slightly the data grouping as expressed by the dispersion parameter k [31], supporting a pre-tectonic magnetization origin (Fig. 4). The mean direction agrees well with other published Permian directions for the Indian craton [28,29,33]. Hence the corresponding pole (Table 2 and Fig. 5) fits the Permian/Triassic segment of the polar wander path from India and Pakistan [26]. The Permian average direction confirms that the Salt Range was once a part of Gondwanaland and was located at a southern latitude of ca. 31° in late Palaeozoic time. The mean declination shows that the beds of

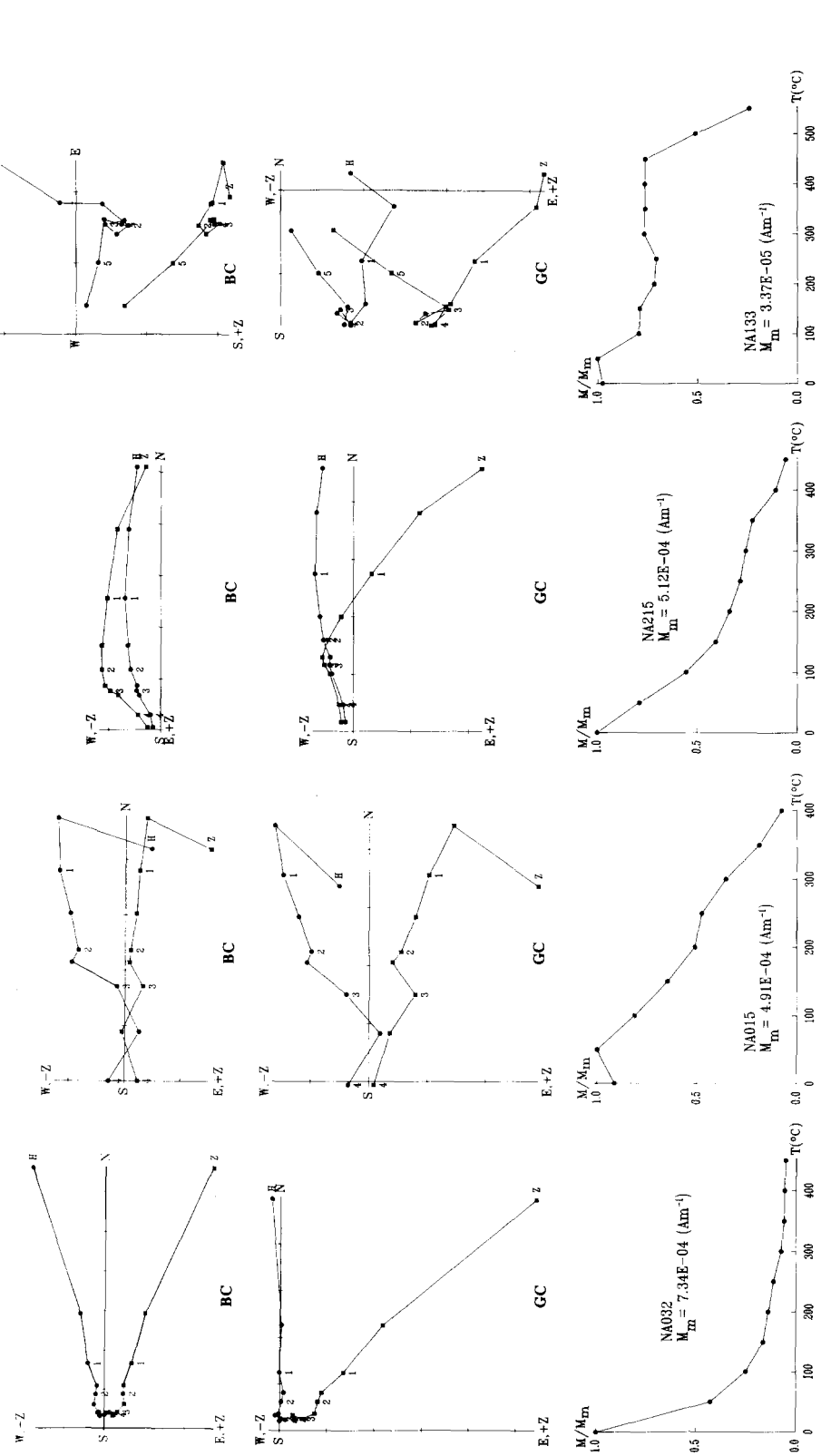


Fig. 3. NRM vectors during thermal demagnetization. Intensities versus temperature [bottom, M_m = maximum magnetization]. *Orthogonal projection of the* NRM horizontal (*H*) and vertical components (*Z*) in geographic coordinates (*GC*) [middle] or after bedding correction (*BC*) [top]; temperatures in °C/100. (a) Sample NA032 with *identifiable present-day component only.* (b) Shallow inclination component of probably Tertiary age predominates in sample NA015. (c) Middle steep NNW upwards directed normal polarity component of probably Permian age observed in sample NA215 after removal of a present-day field component between 300 °C and 450 °C. (d) Reversed polarity of Permian age in sample NA133 between 200 °C and 550 °C. Above 400–450 °C mineral changes with associated new magnetizations and intensity increase often prohibit any further observations of remaining magnetization components. *Therefore demagnetization paths are only shown up to this temperature.*

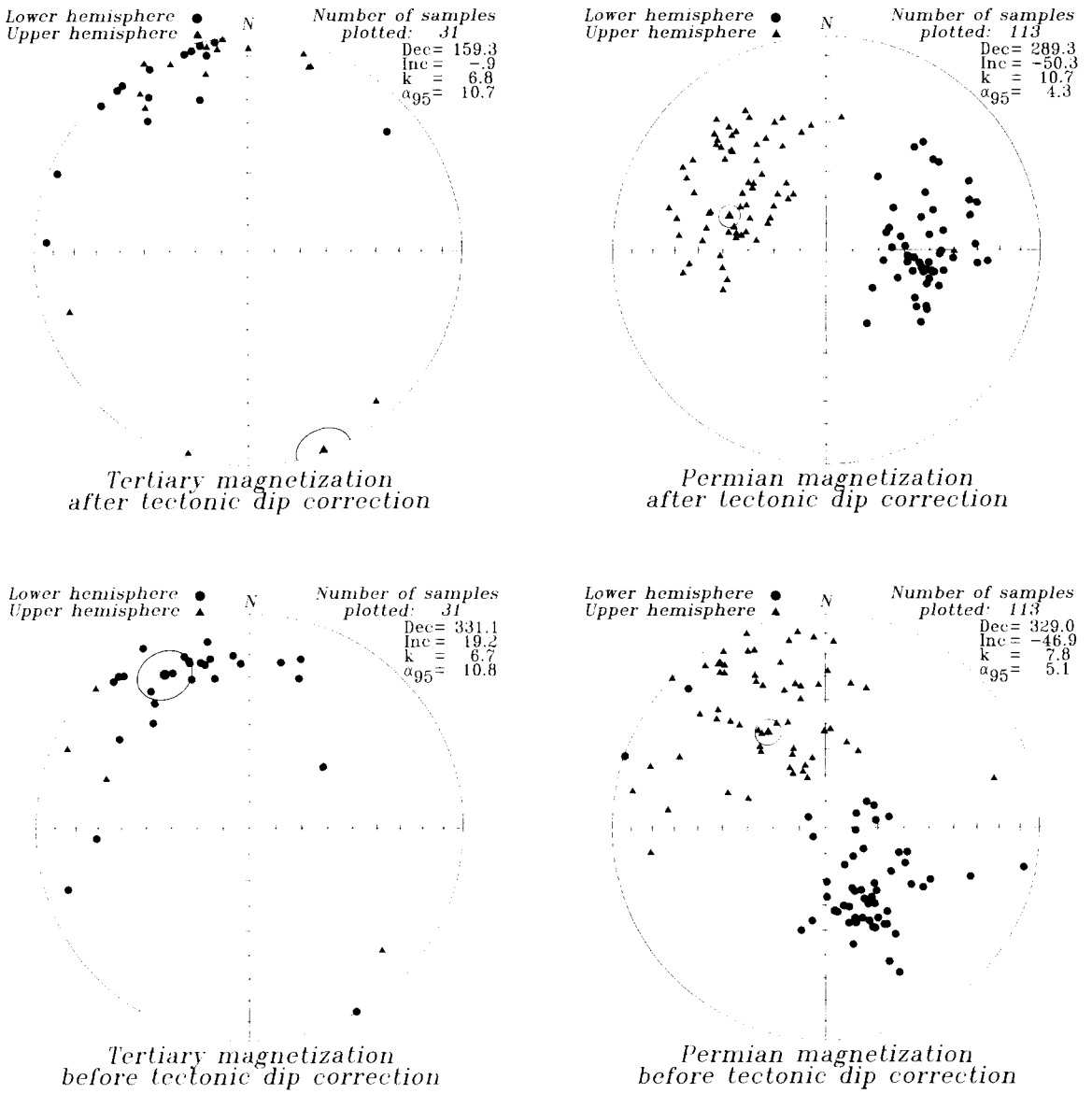


Fig. 4. Stereographic projection of Tertiary and Permian magnetization directions before [bottom] and after tectonic correction [top].

TABLE 2

Magnetization mean directions and palaeopoles after tectonic correction of the Tertiary and Permian magnetizations at Nammal

	Decl.°/Incl.°	N	α°_{95}, k	Palaeopole Lat.N°/Long.E°	δ°_{p95}	δ°_{m95}	A°_{95}
Tertiary	339.3/0.9	31	10.7 6.8	52.4/287.2	5.4	10.7	10.3
Permian	289.3/−50.3	113	4.3 10.7	2.2/125.8	3.8	5.7	4.9

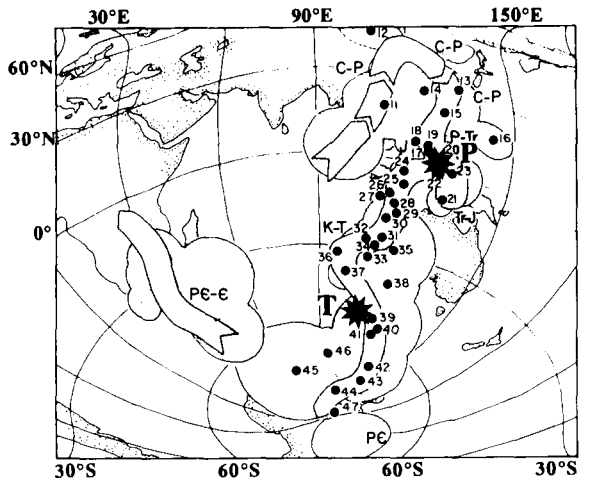


Fig. 5. Apparent polar wander path for Indo-Pakistan modified from [26] and supplemented with the new data from the Nammal gorge (T = Tertiary pole; P = Upper Permian pole). Numbers denote Phanerozoic palaeopoles of the following ages: 11–13 = C–P; 14 = P–Tr; 15 = P_u–Tr_l; 16–17 = P_l; 18–20 = P_u–Tr_l; 21 = Tr_u; 22–23 = J_m; 24–25 = 100–105 Ma; 26–28 = K_{l-u}; 29–33 = K_l, 34 = 65–75 Ma; 35–37 = 60–65 Ma; 38–40 = 55–65 Ma; 41 = 45–55 Ma; 42 = 35–45 Ma; 43 = 25–35 Ma; 44–47 = 5–22.5 Ma.

the Nammal gorge have been rotated anticlockwise by an angle of about 71° relative to the geomagnetic pole, giving only 10°–20° of relative rotation between the Salt Range and the Indian basement since the Permian. This again supports the results of earlier observations [32,33].

Thermal demagnetization was used for the isolation of characteristic NRM components also because acquisition of isothermal remanent magnetization (IRM) frequently indicated the presence of high-coercivity minerals (Fig. 6). Some samples were strongly pre-dominated by goethite with maximum blocking temperatures $\leqslant 50\,^{\circ}C$ (Fig. 6a) and less so by additional haematite (Fig. 6b and c), both minerals carrying present-day field NRM. Microscopic grain structures, intergrowths and replacements in pyrite observed in thin sections suggest that pyrite has been converted to goethite almost completely due to weathering. Haematite is produced at least partly from the goethite by dehydration. Therefore, secondary magnetizations residing in these minerals seem to originate from chemical magnetization processes (cf. [34]). The IRM maximum blocking temperatures of haematite clearly exceed 600 °C. If a sample contains both secondary components, goethite usually carries the present-day field and haematite the Tertiary magnetization. Low-coercivity minerals which shape the main characteristics of many samples (Fig. 6c and d) carry the primary ChRM. They could not be identified under the microscope probably because of their very fine grain size. They often have low maximum blocking temperatures of around 400–450 °C. This low blocking temperature spectrum may explain why the primary magnetization is frequently overprinted by a secondary magnetization. Although Curie temperatures could not be measured because the ferromagnetic mineral content is too small—and enrichment through extraction techniques failed—we propose the presence of very fine grained magnetite or maghemite.

4. Magnetic polarity in the Late Permian and Early Triassic

The Upper Permian magnetization directions of normal and reversed polarity (Fig. 4) which have been established in about half of the Nammal sample collection have been plotted as function of profile depth (Fig. 7). A well-defined polarity zonation is clearly visible. Several broad polarity zones which range in thickness between 10 and 50 m characterize the Wargal and Chhidru formations. These are interrupted at three places by shorter events which at one point (at the 125 m profile depth) are represented by only a single sample. In a 20 m thick zone of the lowermost Chhidru formation, at the 40–60 m profile depth, Permian directions could be detected in only one sample because of a strong Tertiary overprint in this interval.

Using the time scale of Haq and van Eysinga [16] the two formations are interpreted to have been deposited between ca. 261 Ma and 250 Ma ago. This leads to an average sedimentation rate of about 19 m/Ma, assuming constant and continuous deposition. Thus the longest reversed epoch in the middle of the Wargal formation is estimated to have lasted more than 3 Ma, whereas the shortest polarity zone in the upper Wargal covers ca. 75,000 years. We are aware that the deposition rate may vary in the different lithologies: it is a more adequate measure in the limestone sequences than in the sandy beds. Therefore the values given above are rough estimates only.

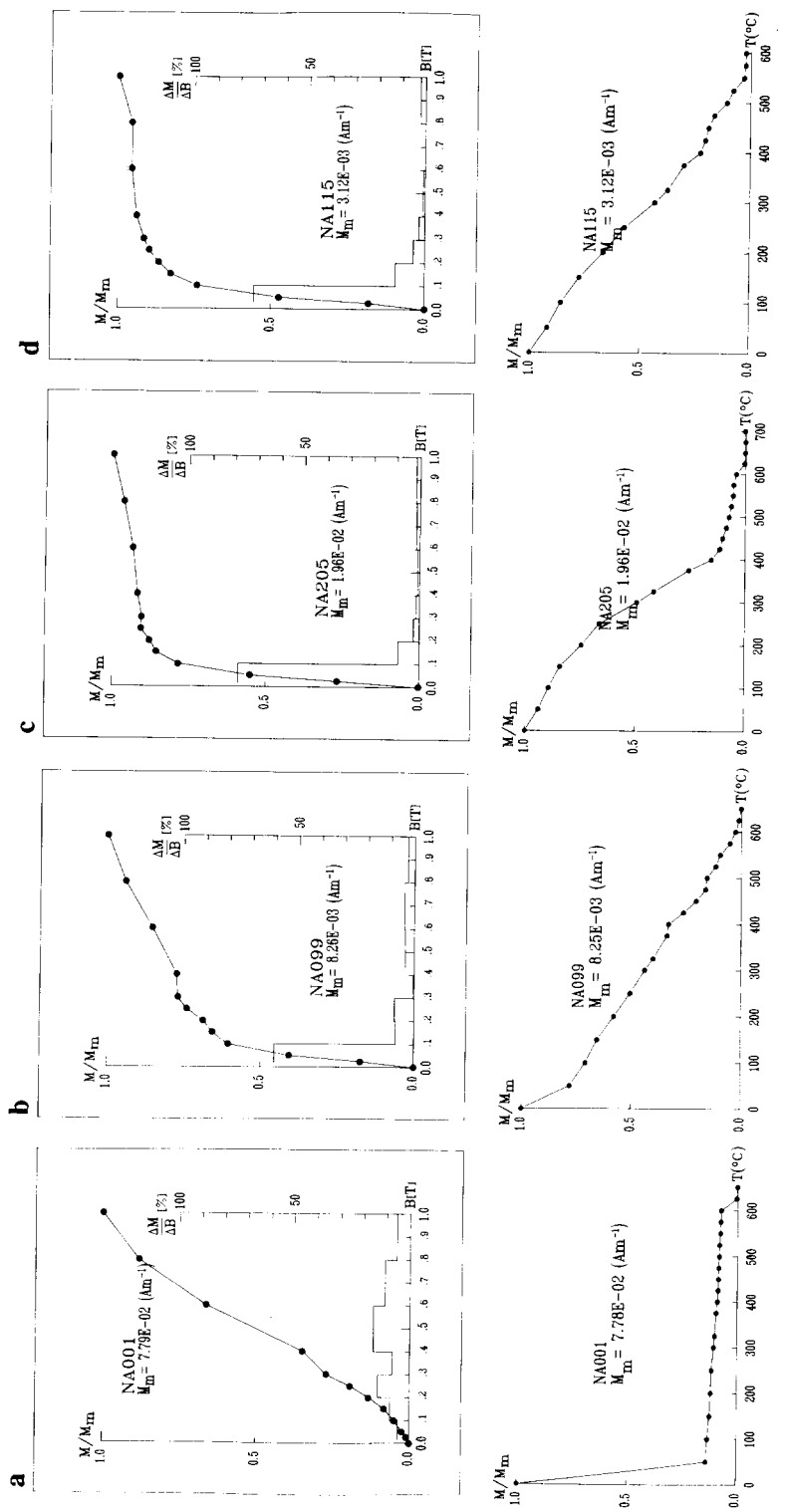


Fig. 6. IRM acquisition in fields up to 1 T and subsequent thermal demagnetization (M_m = maximum magnetization). The samples show very heterogeneous rock magnetic properties. High-coercivity minerals such as goethite (a and b) and haematite (a–c) and low-coercivity minerals (c and d) with low maximum blocking temperatures occur, but are not confined to certain lithologies, i.e. their occurrence cannot be predicted.

The assignment of a specific polarity zone to the P/Tr boundary at Nammal depends very much on correct biostratigraphic dating. According to the results of the PJRG [19] some parts of one ammonite zone of the uppermost Permian (Dorashamian/Changxingian stage in Table 1)

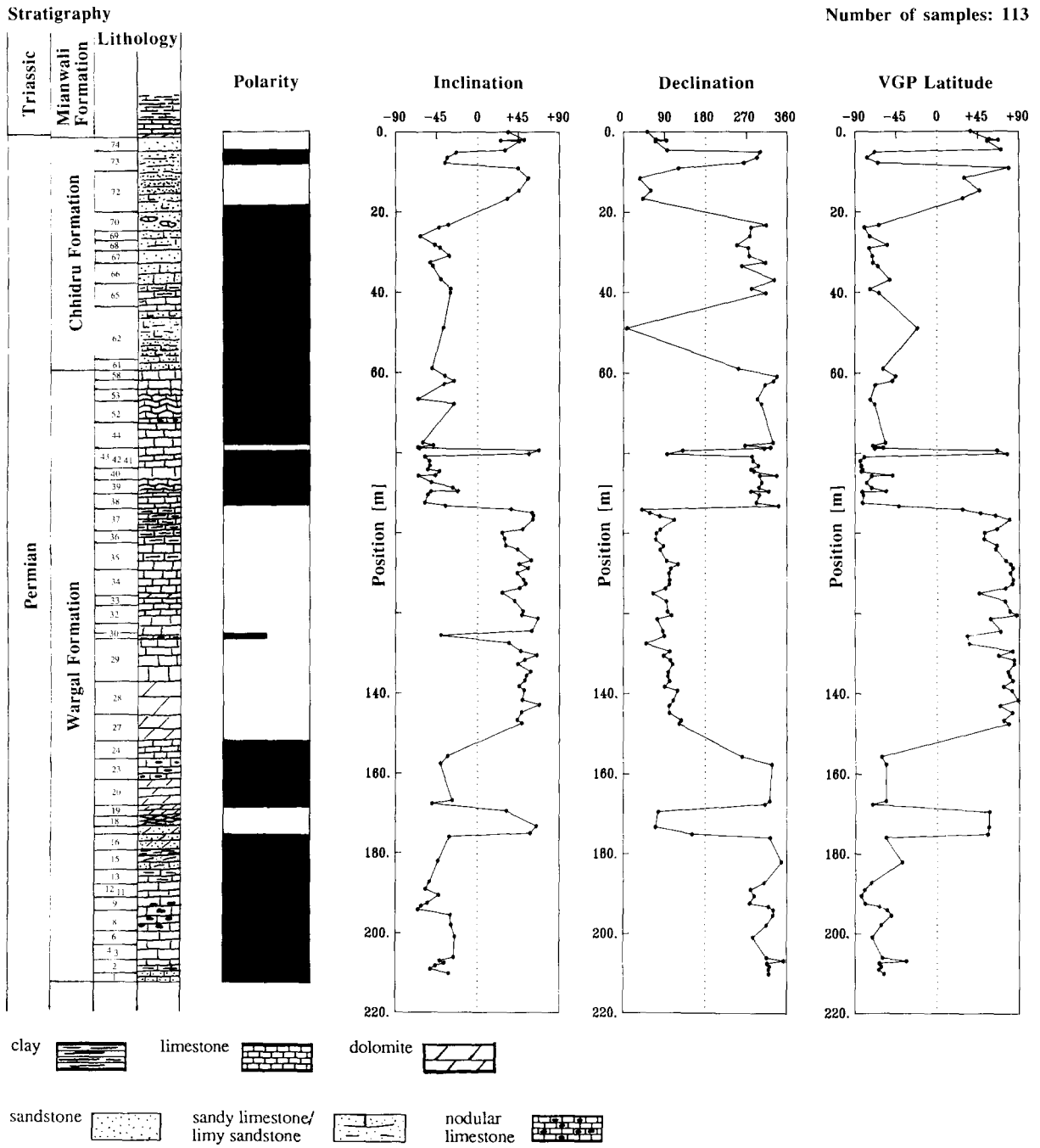


Fig. 7. Magnetostratigraphic section at the Nammal gorge. Black = normal polarity zone; white = reversed; indented polarity zone = reversal which is defined by only one sample. Numbered stratigraphical and lithological units from [19].

are probably missing. The topmost Chhidru formation is thought to be of late Dzhulfian or early Dorashamian age. Only the lowermost Kathwai unit contains clearly identifiable Permian brachiopods of Dorashamian age. This uncertainty raises some doubts on the assignment of the P/Tr boundary to a change from reversed to normal polarity (Fig. 7). There is, however, good agreement with the magnetic polarity pattern of the boundary layers in the stratigraphically most complete profiles from South China [1,2] and the Nammal section. When referring to the stratigraphic correlation of Kummel and Teichert [10], who placed the boundary about half a metre lower, a complete uppermost Permian section could be deduced. The P/Tr boundary would—like in China—fall into the very top part of the underlying reversed polarity zone.

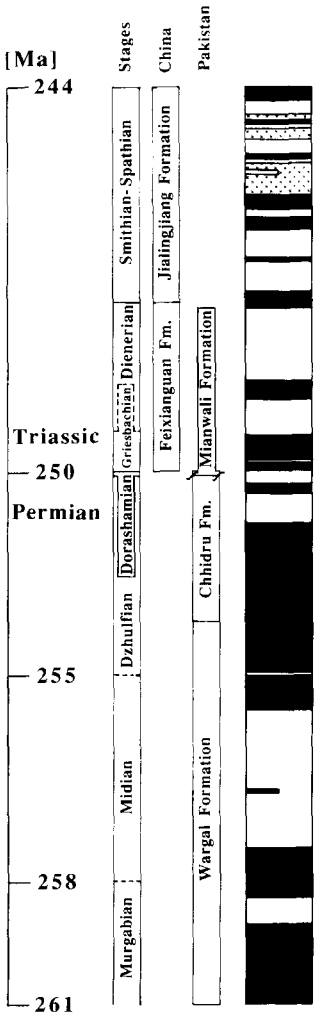


Fig. 8. Composite magnetostratigraphic polarity profile for the Upper Permian (Nammal gorge) to Lower Triassic (South China [1,2]). Formation, stage and age assignment according to [2,5,18,19]. Timescale not linear.

Based on litho- and biostratigraphic evidence, a stratigraphical correlation of regional Upper Permian and lowermost Triassic stages has been attempted (Table 1). Kummel and Teichert [10] correlate the Wargal formation with the Guadalupian of the USA and the Murgabian of Northern Iran and Armenia, whereas the most recent palaeontological examinations [19] place the Wargal/Chhidru formation boundary in the lower Dzhulfian. The well-defined magnetic polarity profile of the Upper Permian sediments in the Nammal gorge contains two distinct normal polarity zones in the lower Wargal formation (Figs. 7 and 8) which extend into the Murgabian. Independent of details of the biostratigraphic zonation these two normal polarity zones of the lower Wargal formation must find correlatives in the Guadalupian and Kazanian or Kungurian. This contradicts the current opinion that rocks of this age (ca. 261 Ma) still belong to the reversed Kiaman hyperzone. The palaeomagnetic reliability of the Chinese Upper Permian sections investigated earlier [1,2] is, unfortunately, very low. The Permian NRM components presented in both these earlier studies are extremely scattered and therefore discarded from further consideration here. Earlier polarity determinations taken from inaccurately dated continental sections in North America and Russia [12,10] placed the end of the Kiaman interval in the upper Tatarian. Our results show that the end of the Kiaman interval and the onset of the Illawarra hyperzone of mixed polarity fall in the early part of the lower Upper Permian. Although sampling was incomplete in the work by Kotlyar et al. [9] and correlation is difficult, the Transcaucasian marine sections also seem to provide evidence of episodes of normal field polarity during the upper Murgabian and Midian.

5. Conclusions

By combining the Lower Triassic magnetostratigraphic sections in South China at Hechuan [2] and Shangsi [1] with the Permian data from the Salt Range in Pakistan we are now able to develop

a more reliable, although still not complete, Upper Permian to Lower Triassic magnetostratigraphic reference polarity profile across the Permian/Triassic boundary (Fig. 8). There are still gaps in the Nammal magnetostratigraphic profile and uncertainties about the exact lithostratigraphic position of this extremely important boundary [23]. More and more evidence accumulates, however, to indicate that the P/Tr boundary is situated close to a change from reversed to normal polarity. The well-defined magnetization directions of the marine sediments of the Nammal gorge show two distinct normal polarity zones in the lower Upper Permian. Therefore, the mixed polarities of the Illawarra hyperzone begin already in the Murgabian (equivalent to the Guadalupian [5] and representing the Kazanian or even Kungurian stages), clearly predating the upper Tatarian. After the end of the Kiaman reversed interval the frequency of polarity changes seems to be rather low in the Upper Permian and of the order of one reversal per million years, resembling the behaviour of the geomagnetic field after the end of the Cretaceous Long Normal Superchron [35]. In the Lower Triassic the frequency seems to double, again similar to the frequency increase observed for the earlier part of the Cenozoic reversal sequence.

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